Evaluating XAI Robustness via Adversaries

Explanation-Guided Attacks for Simulating False Bias

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Motivation

- Machine Learning models are widely deployed in:
  - Finance, healthcare, hiring, law
- These models are often **black-box**
- Need for:
  - Transparency
  - Fairness auditing
- **Post-hoc explainability tools** like:
  - LIME
  - SHAP

  are used to justify model decisions

Core Question

Key Question

Can we trust explanations produced by SHAP and LIME?

- These methods are widely used for:
  - Bias detection
  - Regulatory compliance
- But:
  - Do they truly reflect model behavior?
  - Or can they be manipulated?

Project Idea

False Bias Attack

We construct a model such that:

- Predictions are **correct and fair**
- But SHAP/LIME explanations show:
  - Strong dependence on sensitive features

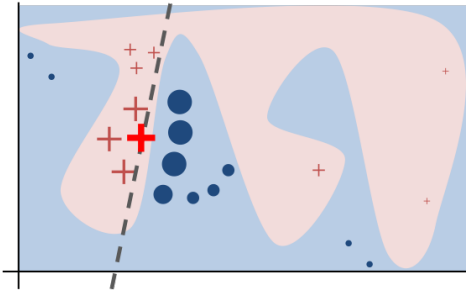
- Goal: **Break trust in post-hoc explanations**
- Show that:
  - Explanations $\neq$ true reasoning

Background: LIME

- Assumption: Model is **locally linear**
- Around a point x, approximate with linear model

$$w^* = \arg\min_w \sum_{i=1}^{N} \pi_x(z_i)\big(f(z_i) - w^T z_i\big)^2$$

- z_i: perturbed samples
- π_x: proximity weighting
- We can also add a regularization term $\Omega(W)$



LIME: Local linear approximation

Background: SHAP

- Based on **Shapley values**
- Measures contribution of each feature

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} (v(S \cup \{i\}) - v(S))$$

- Fair distribution of prediction among features

How SHAP Computes Values

$$v(S) = \mathbb{E}[f(X) \mid X_S = x_S]$$

- Missing features replaced via expectation
- Requires sampling from data distribution

Key Observation

Common Mechanism

Both LIME and SHAP rely on **perturbing inputs**

- Generate synthetic samples near input
- Query model on these samples
- Infer feature importance

This is the core weakness

Data Manifold Insight

- Real data lies on a **low-dimensional manifold**
- Features are **correlated**

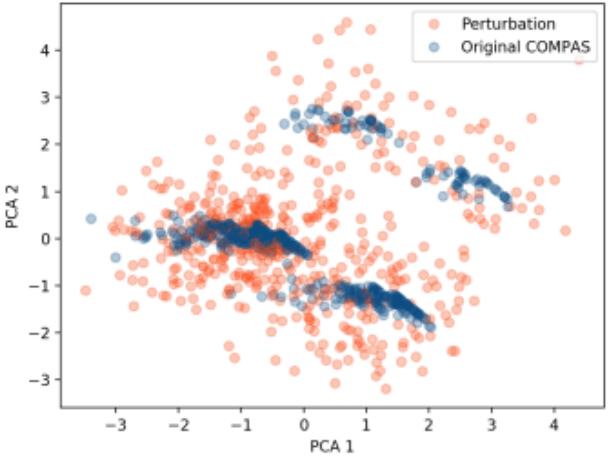
Example:

- House size $\leftrightarrow$ price

Problem:

- LIME/SHAP sample in $\mathbb{R}^d$
- Ignore correlations

LIME perturbation on COMPAS data



Out-of-Distribution (OOD)

$$\inf_{y \in \mathcal{M}} \|z - y\|_2 > \epsilon$$

- Most perturbed samples are **off-manifold**
- Model behavior on these points is:
  - Unconstrained
  - Unreliable

Core Vulnerability

Key Idea

Explanations depend on model behavior on OOD samples

- But model is only trained on real data
- So:
  - We can manipulate behavior on OOD points

Our Approach: Scaffolding Attack

$$F(x) = d(x) f_{\text{fair}}(x) + (1 - d(x)) f_{\text{biased}}(x)$$

- f_{fair}: correct model
- f_{biased}: adversarial model
- $d(x)$: detects OOD vs real input

How Attack Works

- For real inputs:
  - Use f_{fair}
  - Predictions remain correct
- For perturbed inputs:
  - Use f_{biased}
  - Force reliance on sensitive features

Result

Explanations become misleading

Alternative Attack: Gradient Manipulation

$$L' = L - \alpha \|\nabla_{x_{\text{sens}}} L\|_p$$

- Increase gradient along sensitive feature
- LIME/SHAP interpret this as importance

Local vs Global Behavior

Key Idea: Gradients are local

- $\nabla_{x_{\text{sens}}} L$ measures sensitivity **at a point**
- Large gradient $\Rightarrow$ small change in x_{sens} causes large change in prediction
- This does **not** imply global dependence on sensitive features

What the attack does:

- Forces high sensitivity **near training points**
- Model can still ignore sensitive features **elsewhere**

Conclusion:

- Globally fair model
- Locally sensitive behavior

Why LIME / SHAP Get Fooled

How LIME/SHAP work:

- Generate small perturbations around input x
- Fit a local approximation of $f(x)$
- Feature importance $\approx$ local sensitivity

What happens under gradient manipulation:

- Perturbations stay close to training point
- In this region, sensitivity to x_{sens} is artificially high
- Local model assigns high importance to sensitive feature

Result:

- Explanations indicate bias
- But actual predictions remain fair on real data

Expected Outcome

- Model predictions:
  - Accurate
  - Fair
- Explanations:
  - Show **false bias**

x

Evaluation Plan

- Prediction accuracy unchanged
- Measure:
  - Feature importance shift
  - Rank change of sensitive features

Implications

- Explanations can be **manipulated**
- Fair models can be framed as biased
- Risks:
  - Regulatory misuse
  - Corporate attacks

Conclusion

- SHAP and LIME rely on perturbations
- Perturbations are often OOD
- This creates a fundamental vulnerability
- Our work demonstrates:
    - Explanations can be adversarially controlled

Questions?